Samarth Parekh

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Education

University of Maryland College Park, MD
Bachelor of Science in Computer Science, Minor in Mathematics December 2026
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Computer Systems, Programming Languages, Quantum Computing, Quantum Hardware, Bioinformatics, Human-Robot Interactions, Linear Algebra, Abstract Algebra, Calculus III, Discrete Structures, Probability & Statistics

Experience

Software Engineer Intern May 2026 - August 2026
Stripe | *Ruby, TypeScript, React.js, Claude Code, Codex* Seattle, WA
- Engineered an automated testing framework in Ruby that validated over 24,000 unique combinations of merchant states and dashboard surfaces, preventing a recurring class of critical production incidents that had blocked dozens of global merchants from their finances
- Architected and led the consolidation of 29 dynamic UI widgets into a unified React/TypeScript library, enabling seamless code reuse and a consistent user experience across multiple high-traffic dashboard surfaces
- Designed and implemented a secure internal API to facilitate protected document retrieval, migrating 5+ legacy endpoints into a shared frontend package while maintaining strict security and permission policies for sensitive merchant data

Software Engineer Intern June 2025 - August 2025
StoneX | *Apex, Salesforce, Python, Microsoft Power Platform* New York, NY
- Reduced post-ingestion field updates by 40% by designing and developing REST API enhancements that integrated external applications with Salesforce, minimizing manual corrections and increasing CRM data reliability
- Cut manual ticket handling by 65% by developing an Outlook automation tool with Power Automate and a custom text-classification model, accelerating ticket triage and increasing support throughput
- Increased CRM operational efficiency and data accuracy across multiple regions by collaborating with cross-functional global teams to optimize BAU workflows and user interfaces, resulting in streamlined procedures and boosted user adoption

Quantum Machine Learning Engineer February 2025 – May 2025
IonQ | *Python, Pennylane* College Park, MD
- Benchmarked classical vs. hybrid quantum-classical models on MNIST by executing circuits on IonQ Aria and Forte quantum computers, delivering insights for model selection on resource-constrained quantum hardware
- Increased efficiency of quantum SVMs and GANs on limited-qubit QPUs by designing optimized feature-reduction pipelines, reducing qubit requirements while improving classification accuracy
- Investigated quantum-enhanced approaches for analyzing neural spike trains, studying higher-order correlations and connectivity models to evaluate potential quantum advantages in neuroscience applications

Software Engineer Intern June 2024 – May 2025
US Food and Drug Administration | *Python, Dash, TypeScript* College Park, MD
- Lowered manual test-data preparation time by 30% by designing and implementing a Python/Dash synthetic-data generator, accelerating testing cycles and improving developer productivity
- Enabled traceability adoption across 10+ international organizations by authoring comprehensive system and API documentation for an open-source tool, reducing manual effort and easing integration
- Increased tech-enabled food traceability initiative visibility by presenting a poster to 200+ attendees at the FDA Foods Program Regulatory Science Conference, sparking new collaboration opportunities

Software Engineer - Tech Lead September 2024 – December 2024
Warriors Legacy Care | *React Native, Flask, PostgreSQL, AWS, Google Gemini, Figma* College Park, MD
- Secured $5M in funding from government grants and stakeholders by leading a 10-member engineering team to design and launch a React Native app, providing veterans with access to care facilities, mental health resources, and secure messaging
- Improved veteran job-search support by developing an AI-powered resume reviewer with Google Gemini and a real-time job-posting scraper, delivering tailored recommendations
- Enhanced real-time service delivery for veterans by integrating WebSockets, geolocation, and REST APIs to enable live messaging, dynamic provider mapping, and appointment scheduling, reducing friction in accessing care

Projects

Matchrimoney | **Wedding Vendor Cost-Sharing** | *React.js, Python, Flask, OpenAI* View Code
- Enabled couples to cut wedding expenses in a $70B+ market by building a production-ready platform with vendor matching and cost-sharing features, helping users save money and simplify planning
- Ensured a reliable client onboarding and user experience by validating data integrity and designing the system for pilot users

EcoNavix | **AI-Driven Route Optimization** | *React.js, OpenAI Node.js, Tailwind, PostgreSQL* View Code
- Reduced fleet carbon emissions by 35% by integrating traffic, weather, and emissions APIs into real-time dynamic routing
- Accelerated logistics decision-making by engineering a scalable backend with REST APIs that delivered sustainability insights to client dashboards

Technical Skills

Languages: Python, Java, Apex, TypeScript, JavaScript, HTML/CSS, C, MATLAB, SQL
Libraries and Frameworks: React.js, React Native, Node.js, Express.js, Stripe, Clerk, JUnit
Developer Tools: AWS, Salesforce, Git, GitHub, Postman, Firebase, MongoDB, Google Suite, Microsoft Platforms, Jira, Linux
Certifications: AWS Cloud Practitioner, AWS AI Practitioner